



M003 Guide — Counting (x, y, z) · Beginner

Topic: Before 3D · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 20 minutes

Pixel art used **two** numbers. A 3D build needs **three**. They always come in the same order:

1st = X

right → left
starts at 0

2nd = Y

down → up
starts at 0

3rd = Z

back → forward
starts at 0

Count x, then y, then z. Always that order.

● Stand on your **home spot** (red block) every run. Move your feet, move your build.

1 · Build Your Grid 🎨

Make a striped ruler for each axis, so you can **see** the numbers.



Stand on your home spot. Run this to lay the white floor and the yellow lines:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(0, -1, 0),
        pos(15, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(5, -1, 0),
        pos(5, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(10, -1, 0),
        pos(10, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(0, -1, 5),
        pos(15, -1, 5),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(0, -1, 10),
        pos(15, -1, 10),
        FillOperation.REPLACE)
    player.on_chat("grid", on_run)
```

Now place the three rulers by hand — **yellow, black, yellow, black** all the way. All three start on the block under your feet:

- **X ruler** — one block at a time, **to your left**.
- **Z ruler** — one block at a time, **forward**.
- **Y ruler** — a pillar straight **up**.

  The stripes are there so you can count without guessing.

2 · Start at the Corner ●

The three rulers all meet on **one block**. That block is your home spot.

Its coordinate is **(0, 0, 0)**. Every count starts there.

From the corner	You get
stand still	(0, 0, 0)
1 step left	(1, 0, 0)
1 block up	(0, 1, 0)
1 step forward	(0, 0, 1)

x, y and z all start at which number? Circle one: 0 · 1

Where is (0, 0, 0)? Circle one: the corner · one block up

3 · The Three Rulers

Each ruler starts at the corner and counts **0, 1, 2, 3 ...**



X — right → left



Y — down → up



Z — back → forward

```
place red block at (1, 0, 0)  # 1 step left
place red block at (0, 1, 0)  # 1 block up
place red block at (0, 0, 1)  # 1 step forward
```

Which ruler goes to your left? Circle one: $x \cdot y \cdot z$

A block at $y = 3$ is ...? Circle one: on the ground · 3 blocks up

Which ruler goes forward? Circle one: $x \cdot y \cdot z$

4 · Put Them Together

Read **(5, 2, 3)** in three steps:

Step	Number	Ruler	Do this
1	$x = 5$	right → left	go 5 to your left
2	$y = 2$	down → up	go 2 up (the ground is 0)
3	$z = 3$	back → forward	go 3 forward

Which coordinate is on the ground? Circle one: $(2, 1, 2) \cdot (2, 0, 2)$

These two blocks are stacked. Which number grew?

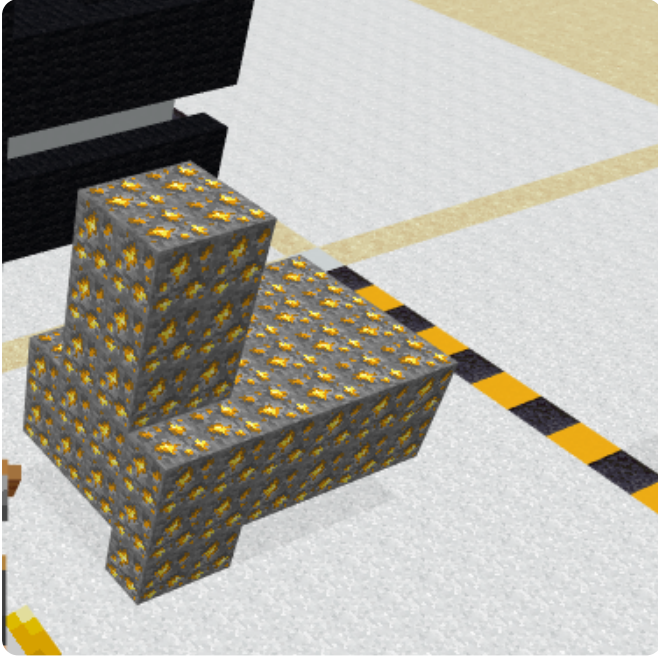
```
place red block at (7, 1, 2)
place red block at (7, 2, 2)
```

Circle one: $x \cdot y \cdot z$

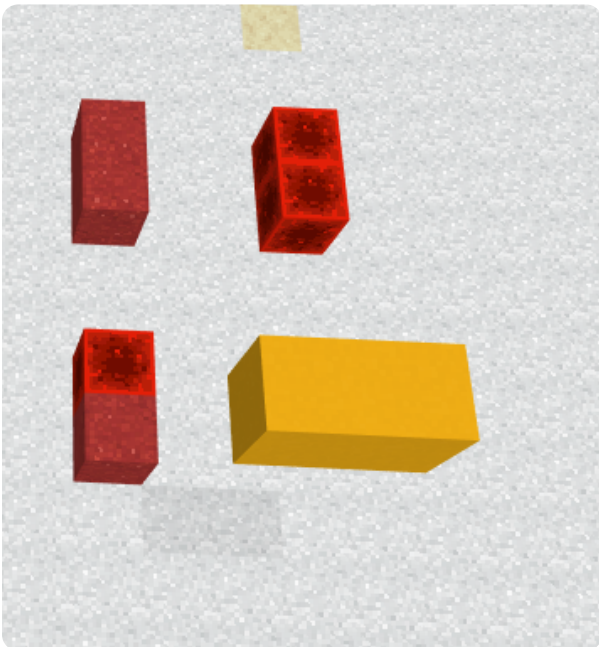
5 · Show Your Work 📷 🎥

Warm-up: stand on your home spot and place one block at **(3, 0, 3)**. Check it against your rulers.

Open the world. Part of the shape is **already built** for you:



The rest of the blocks are **missing**. Here they are, spread apart so you can see each one:



Finish the shape. Count x, then y, then z for every block you place.

Write each block's coordinate here before you place it.



The corner block is **(0, 0, 0)**. Use the same blocks you see in the world.

Record **one video** (a phone is fine). Show two things:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Count one block out loud — x, then y, then z.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Submit

Send this worksheet + your video to teacher on KakaoTalk.